Problem Set 3

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Invalid Date

Question 1

```r
main_data <- read_csv(here("Y2S1/IO/PS3/Data/prod_level_data.csv"))
```

```
Rows: 71 Columns: 21
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (21): market, x1, x2, x3, x4, p1, p2, p3, p4, ave_dist1, ave_dist2, ave_...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
if (all(c("x", "p", "ave_dist", "s", "mc") %in% names(main_data))) {
  main_long <- main_data
} else {
  main_long <- main_data %>%
    pivot_longer(
      cols = matches("^(x|p|ave_dist|s|mc)\\d+$"),
      names_to = c(".value", "prod"),
      names_pattern = "(.*?)(\\d+)$"
    ) %>%
    mutate(prod = as.integer(prod)) %>%
    arrange(market, prod)
}

head(main_long)
```

```
# A tibble: 6 x 7
  market  prod     x     p ave_dist     s    mc
   <dbl> <int> <dbl> <dbl>    <dbl> <dbl> <dbl>
1      1     1 2.02   8.71   0.0875 0.48  0.768
2      1     2 0.252  5.20   0.0865 0.098 0.824
3      1     3 3.47   5.10   0.0684 0.163 0.348
4      1     4 0.306  5.03   0.0474 0.12  0.464
5      2     1 0.942  5.43   0.553  0.143 0.721
6      2     2 0.909  4.80   0.966  0.039 0.672
```

```r
describe(main_long)
```

```
         vars   n  mean    sd median trimmed   mad  min   max range  skew
market      1 284 36.00 20.53  36.00   36.00 26.69 1.00 71.00 70.00  0.00
prod        2 284  2.50  1.12   2.50    2.50  1.48 1.00  4.00  3.00  0.00
x           3 284  2.57  1.48   2.55    2.58  2.00 0.03  4.99  4.96 -0.05
p           4 284  5.46  0.91   5.22    5.33  0.60 4.17  9.56  5.39  1.76
ave_dist    5 284  0.84  0.90   0.55    0.69  0.47 0.01  6.54  6.53  3.21
s           6 284  0.17  0.12   0.15    0.16  0.10 0.01  0.56  0.56  1.10
mc          7 284  0.52  0.29   0.55    0.52  0.35 0.00  1.00  0.99 -0.09
         kurtosis   se
market      -1.21 1.22
prod        -1.37 0.07
x           -1.28 0.09
p            3.87 0.05
ave_dist    14.91 0.05
s            0.73 0.01
mc          -1.20 0.02
```

```r
stat_tab <- main_long %>%
  select(x, p, ave_dist, s, mc) %>%
  pivot_longer(
    everything(),
    names_to = "variable",
    values_to = "value"
  ) %>%
  group_by(variable) %>%
  summarize(
    N = n(),
    Mean = mean(value),
    SD = sd(value),
```

```r
    Min = min(value),
    P25 = quantile(value, 0.25),
    Median = median(value),
    P75 = quantile(value, 0.75),
    Max = max(value),
    .groups = "drop"
  )

kable(stat_tab,
      digits = 3,
      caption = "Table of Summary Statistics",)
```

Table 1: Table of Summary Statistics

variable	N	Mean	SD	Min	P25	Median	P75	Max
ave_dist	284	0.845	0.899	0.014	0.343	0.546	1.159	6.543
mc	284	0.516	0.289	0.004	0.273	0.549	0.765	0.996
p	284	5.460	0.906	4.168	4.861	5.224	5.755	9.557
s	284	0.174	0.120	0.007	0.086	0.148	0.224	0.564
x	284	2.568	1.478	0.030	1.222	2.548	3.934	4.993

```r
main_graph_pxs <- main_long %>%
  ggplot(aes(x = p, y = s)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    title = "Price vs. Market Share by Product",
    x = "Market Share (s)",
    y = "Price (p)"
  ) +
  theme_minimal()
ggsave(main_graph_pxs,
        filename = here("Y2S1/IO/PS3/Outputs/price_vs_share.pdf"),
        width = 6, height = 4)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```r
main_graph_xxs <- main_long %>%
  ggplot(aes(x = x, y = s)) +
  geom_point() +
```

```r
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    title = "Rating vs. Market Share by Product",
    x = "Rating (x)",
    y = "Market Share (s)"
  ) +
  theme_minimal()
ggsave(main_graph_xxs,
        filename = here("Y2S1/IO/PS3/Outputs/rating_vs_share.pdf"),
        width = 6, height = 4)
```

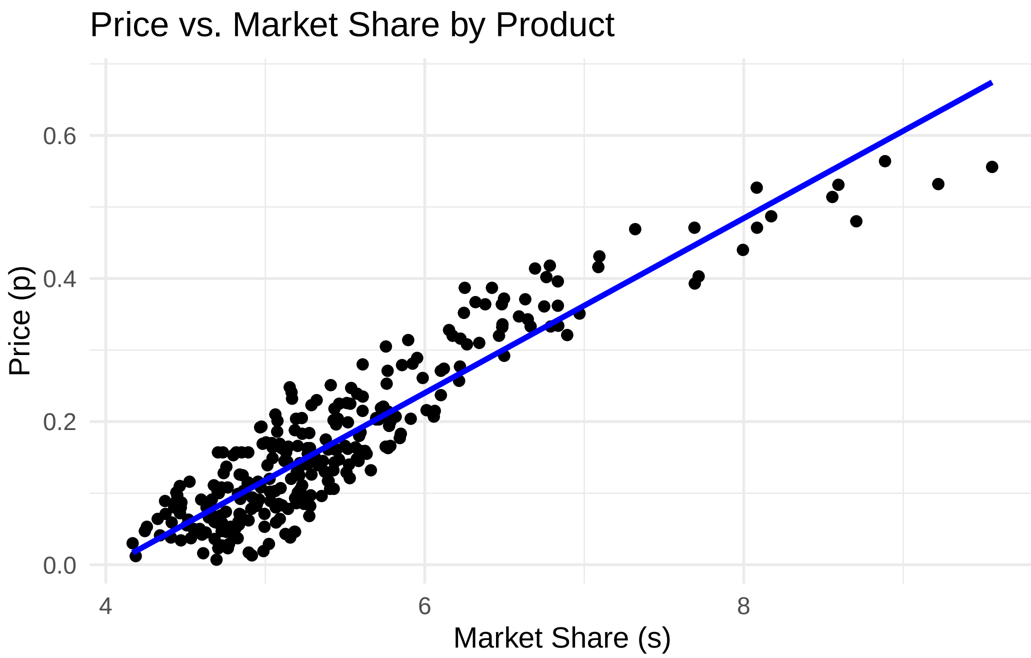
```
`geom_smooth()` using formula = 'y ~ x'
```

```r
main_graph_sxad <- main_long %>%
  ggplot(aes(x = ave_dist, y = s)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    title = "Average Distance vs. Market Share by Product",
    x = "Average Distance (ave_dist)",
    y = "Market Share (s)"
  ) +
  theme_minimal()
ggsave(main_graph_sxad,
        filename = here("Y2S1/IO/PS3/Outputs/distance_vs_share.pdf"),
        width = 6, height = 4)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

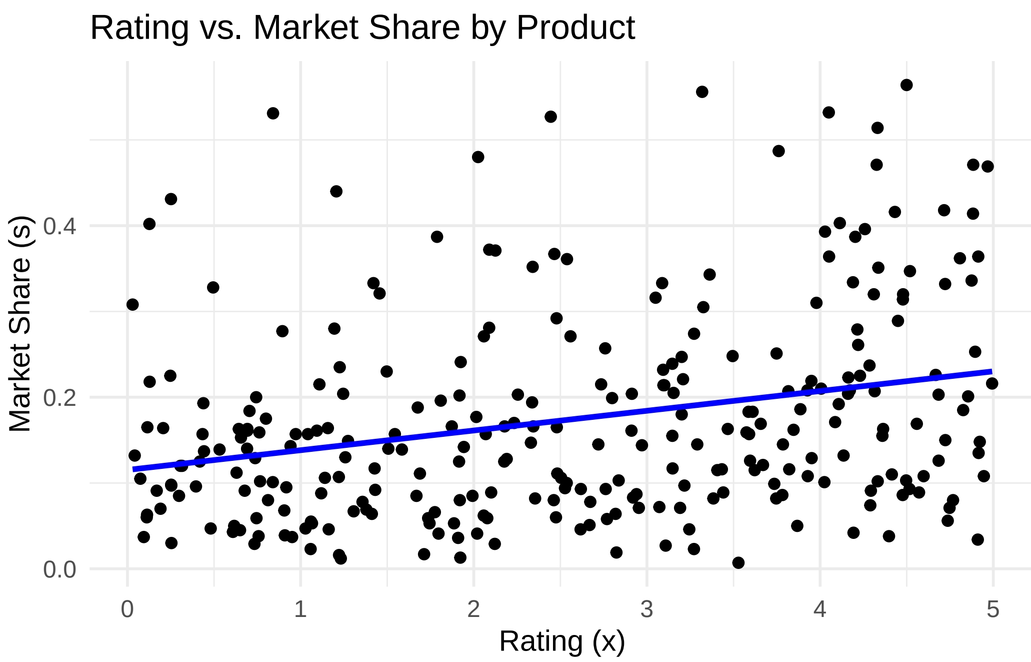
```r
main_graph_pxs
```

```
`geom_smooth()` using formula = 'y ~ x'
```



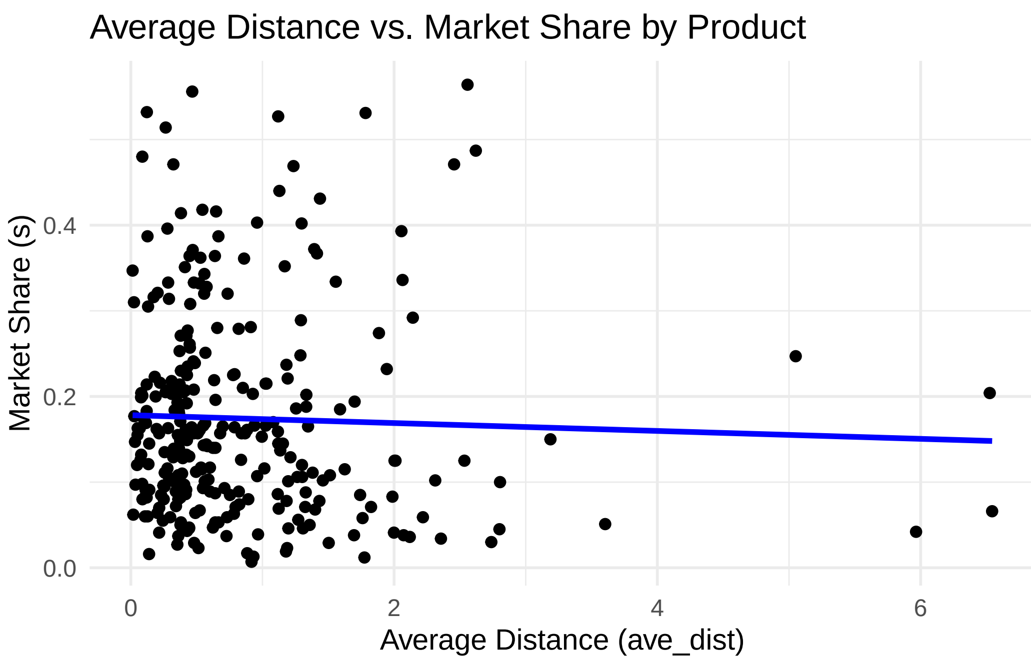

```
main_graph_xxs
```

```
`geom_smooth()` using formula = 'y ~ x'
```



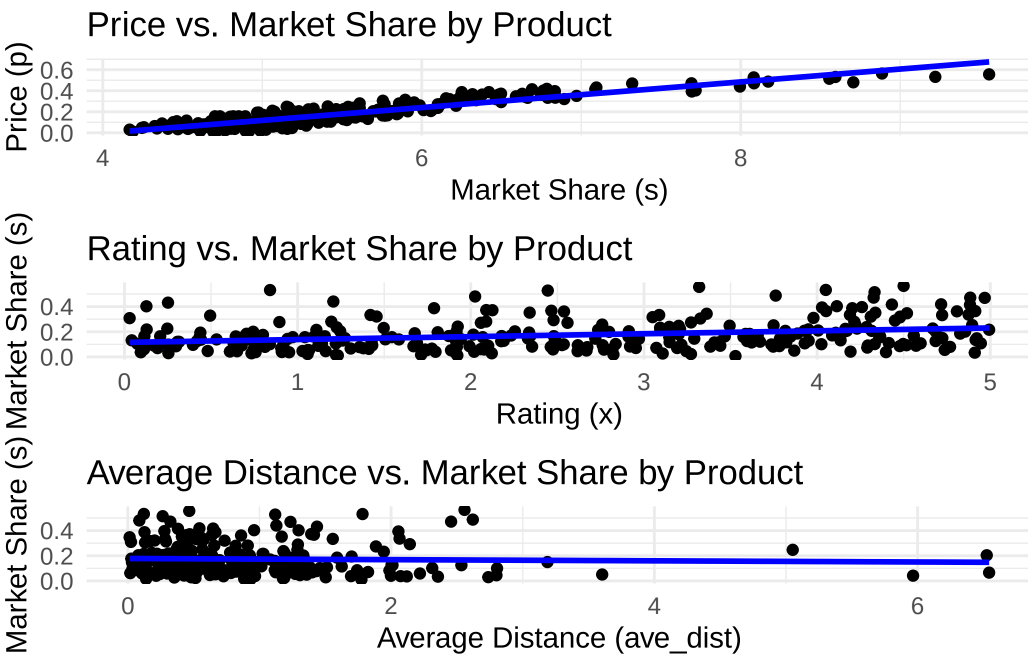
```r
main_graph_sxad
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```r
grid.arrange(main_graph_pxs, main_graph_xxs, main_graph_sxad, ncol = 1)
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



Price is positively correlated with market share, with price increasing with share for all products. For Yelp rating, there is no discernable correlation. For distance, there is a slightly negative relationship, though it appears to be as good as flat. All make sense to me, as price is directly manipulable as market power increases, while distance may be harder to control due to real estate effects, and ratings are almost fully out of control.

Question 2

Part 1

```
logit_data <- main_long %>%
  group_by(market) %>%
  mutate(
    s_out = 1 - sum(s),
    logit_s = log(s) - log(s_out)
  ) %>%
  ungroup()

mnl1 <- feols(
  logit_s ~ x + p,
  data = logit_data
```

```r
)
summary(mnl1)
```

```
OLS estimation, Dep. Var.: logit_s
Observations: 284
Standard-errors: IID
             Estimate Std. Error   t value  Pr(>|t|)
(Intercept) -4.744482   0.199805 -23.74556 < 2.2e-16 ***
x            0.059056   0.022765   2.59414 0.0099801 **
p            0.701882   0.037148  18.89406 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.546478   Adj. R2: 0.591078
```

Part 2

```r
logit_data <- logit_data %>%
  group_by(market) %>%
  mutate(
    zp = mc
  ) %>%
  ungroup()

mnl2 <- ivreg(
  logit_s ~ x + p | x + zp,
  data = logit_data
)
summary(mnl2)
```

```
Call:
ivreg(formula = logit_s ~ x + p | x + zp, data = logit_data)

Residuals:
     Min       1Q   Median       3Q      Max
-3.11303 -0.87252 -0.03998  0.63885  4.13642

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept)  1.70956    2.22696   0.768  0.44333
x            0.24623    0.08168   3.015  0.00281 **
p           -0.56814    0.43724  -1.299  0.19488
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.248 on 281 degrees of freedom
Multiple R-Squared: -1.095, Adjusted R-squared: -1.11
Wald test: 6.085 on 2 and 281 DF,  p-value: 0.002588
```

Part 3

```r
logit_data <- logit_data %>%
  group_by(prod, market) %>%
  mutate(
    ad = ave_dist
  ) %>%
  ungroup()

mnl3 <- ivreg(
  logit_s ~ x + p + ad | x + ad + zp,
  data = logit_data
)
summary(mnl3)
```

```
Call:
ivreg(formula = logit_s ~ x + p + ad | x + ad + zp, data = logit_data)

Residuals:
     Min       1Q   Median       3Q      Max
-3.11693 -0.86123 -0.07583  0.65975  4.13945

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.90340    2.24620   0.847  0.39750
x            0.24942    0.08221   3.034  0.00264 **
p           -0.58279    0.44010  -1.324  0.18651
ad          -0.14442    0.08312  -1.737  0.08341 .
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.255 on 280 degrees of freedom
Multiple R-Squared: -1.111, Adjusted R-squared: -1.134
Wald test: 4.968 on 3 and 280 DF,  p-value: 0.002249
```

```r
stargazer(mnl1, mnl2, mnl3,
          type = "text",
          dep.var.labels = "Logit Market Share",
          column.labels = c("No Instrument", "Instrument for Price",
                            "Instrument for Price and Distance"),
          covariate.labels = c("Rating (x)", "Price (p)",
                               "Average Distance (ad)"),
          stars = c(0.01, 0.05, 0.1),
          digits = 3,
          out = here("Y2S1/IO/PS3/Outputs/reg_results.txt"))
```

```
% Error: Unrecognized object type.
```

In the first model, we see a clear misspecification, since price's coefficient is positive. However, upon instrumenting for price, we flip the sign and get a negative coefficient. But price is not not statistically significant, indicating a weak instrument. Finally, adding distance as an instrument, we see that both price and distance have negative signs, which is to be expected, but neither is statistically significant.

If the true values are the values set for simulation as laid out in part 1, price is biased negatively, distance is slightly biased negatively, and characteristics are slightly biased positively. Thus, all parameters estimated through multinomial logit are overstating the effect of these variables when in the case of complete information.

Question 3

```r
search_dat <- read_csv(here("Y2S1/IO/PS3/Data/search_set_data.csv"))
```

```
Rows: 1136 Columns: 5
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (5): market, prod1, prod2, prod3, prod4
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
s_mat <- search_dat %>%
  filter(market == min(market, na.rm = TRUE)) %>%
  select(prod1, prod2, prod3, prod4) %>%
  as.matrix()

stopifnot(nrow(s_mat) == 16, ncol(s_mat) == 4)

q4_df <- main_long %>%
  mutate(prod = as.integer(prod)) %>%
  arrange(market, prod)

q4_df <- q4_df %>%
  group_by(market) %>%
  filter(s > 0) %>%
  ungroup()
```

```r
n <- nrow(q4_df)
Z <- model.matrix(~ x + mc + ave_dist, data = q4_df)
W <- solve(crossprod(Z) / n)

eps <- 1e-12

delta_0 <- q4_df %>%
  group_by(market) %>%
  mutate(
    s0 = 1 - sum(s),
    delta_0 = log(s) - log(s0)
  ) %>%
  ungroup() %>%
  pull(delta_0)

idx_vec <- split(seq_len(n), q4_df$market)

pi_func <- function(mu, ave_dist, eps = 1e-12) {
  pi <- plogis(10 + (mu * ave_dist))
  pmin(pmax(pi, eps), 1 - eps)
}
```

```r
predict_share_func <- function(delta, pi, s_mat) {
  fixer <- max(0, max(delta))
  exp_fixed <- exp(delta - fixer)
  denom <- exp(-fixer) + as.vector(s_mat %*% exp_fixed)

  logP <- as.vector(s_mat %*% log(pi) + (1 - s_mat) %*% log(1 - pi))
  P <- exp(logP)

  con_P <- sweep(s_mat, 2, exp_fixed, `*`)
  con_P <- con_P / denom
  colSums(con_P * P)
}

con_map <- function(mu, delta_start, df, idx_vec, s_mat,
                    tol = 1e-10, max_iter = 1000) {

  eps <- 1e-12

  s_obs <- pmax(df$s, eps)
  pi_all <- pi_func(mu, df$ave_dist)

  delta <- delta_start

  for (i in seq_len(max_iter)) {
    s_hat <- numeric(length(delta))
    for (g in seq_along(idx_vec)) {
      idx <- idx_vec[[g]]
      s_hat[idx] <- predict_share_func(delta[idx], pi_all[idx], s_mat)
    }
    s_hat <- pmax(s_hat, eps)

    delta_new <- delta + log(s_obs) - log(pmax(s_hat, 1e-12))
    diff <- max(abs(delta_new - delta))

    delta <- delta_new
    if (is.finite(diff) && diff < tol) {
      return(list(delta = delta, converged = TRUE, iter = i, diff = diff))
    }
  }
  list(delta = delta, converged = FALSE, iter = max_iter, diff = diff)
}
delta_last <- delta_0
```

```r
gmm_exog <- function(theta) {
  beta0 <- theta[1]
  beta1 <- theta[2]
  alpha <- theta[3]
  mu <- theta[4]

  cm <- con_map(
    mu,
    delta_last,
    q4_df,
    idx_vec,
    s_mat, tol = 1e-12, max_iter = 1000)

  if (!cm$converged || any(!is.finite(cm$delta)) || max(abs(cm$delta)) > 1e3) {
    return(1e12)
  }

  delta_last <<- cm$delta

  xi <- cm$delta - (beta0 + beta1 * q4_df$x + alpha * q4_df$p)
  Q <- crossprod(Z, xi) / n
  obj <- as.numeric(t(Q) %*% W %*% Q)

  if (!is.finite(obj)) obj <- 1e12
  obj
}

if (exists("m3")) {
  b <- coef(m3)
  theta_start <- c(
                  beta0 = unname(b["(Intercept)"]),
                  beta1 = unname(b["x"]),
                  alpha = unname(b["p"]),
                  mu = -1.0)
} else {
  theta_start <- c(
    beta0 = 1.0,
    beta1 = 0.2,
    alpha = -0.25,
    mu = 0
  )
}
```

```r
q4_df %>%
  group_by(market) %>%
  summarise(
    sum_s = sum(s),
    min_s = min(s),
    .groups = "drop"
  ) %>%
  arrange(desc(sum_s))
```

```
# A tibble: 71 x 3
   market sum_s min_s
    <dbl> <dbl> <dbl>
 1     33 0.888 0.034
 2     36 0.866 0.062
 3      1 0.861 0.098
 4     29 0.84  0.059
 5     38 0.838 0.037
 6     45 0.833 0.068
 7     69 0.828 0.007
 8      9 0.824 0.023
 9     58 0.822 0.016
10     51 0.799 0.012
# i 61 more rows
```

```r
set.seed(0219)
gmm_opt <- optim(
  par = theta_start,
  fn = gmm_exog,
  method = "Nelder-Mead",
  control = list(maxit = 5000, reltol = 1e-8)
)

theta_hat <- gmm_opt$par
theta_hat
```

```
     beta0      beta1      alpha         mu
 1.5832659  0.2425248 -0.5432585 -0.1622090
```

```r
gmm_opt$value
```

```
[1] 0.01673846
```

```r
gmm_opt$convergence
```

```
[1] 0
```

```r
resolve <- con_map(
  theta_hat["mu"],
  delta_0,
  q4_df,
  idx_vec,
  s_mat, tol = 1e-10, max_iter = 1000
)

est_tab <- tibble(
  term = c("Intercept", "Rating Coefficient", "Price Coefficient", "Mu"),
  estimate = c(theta_hat["beta0"],
               theta_hat["beta1"],
               theta_hat["alpha"],
               theta_hat["mu"]),
  true = c(1, 0.2, -0.25, NA_real_)
)

kable(est_tab, digits = 3, caption = "GMM Estimates vs. True Values")
```

Table 2: GMM Estimates vs. True Values

term	estimate	true
Intercept	1.583	1.00
Rating Coefficient	0.243	0.20
Price Coefficient	-0.543	-0.25
Mu	-0.162	NA

Question 4

```r
loc_dat <- read_csv(here("Y2S1/IO/PS3/Data/distance.csv"))

q5_df <- main_long %>%
```

```r
  arrange(market, prod)

S_temp <- search_dat %>%
  filter(market == min(market, na.rm = TRUE)) %>%
  select(prod1, prod2, prod3, prod4) %>%
  as.matrix()

S <- S_temp[rowSums(S_temp) > 0, , drop = FALSE]
stopifnot(ncol(S) == 4)

loc_list <- split(loc_dat, loc_dat$market)
loc_list <- lapply(loc_list, function(ave_dist) {
  as.matrix(ave_dist[, c("dist1", "dist2", "dist3", "dist4")])
})

n <- nrow(q5_df)
idx_vec <- split(seq_len(n), q5_df$market)

Z <- model.matrix(~ x + mc + ave_dist, data = q5_df)
W <- solve(crossprod(Z) / n)

delta_0 <- q5_df %>%
  group_by(market) %>%
  mutate(
    s0 = 1 - sum(s),
    delta_0 = log(s) - log(s0)
  ) %>%
  ungroup() %>%
  pull(delta_0)
```

```r
predict_share_endog <- function(delta, gamma, D, S) {
  K <- nrow(S)
  N <- nrow(D)

  m <- max(c(0, delta))
  exp_fixed <- exp(delta - m)
  sum_exp <- as.vector(S %*% exp_fixed)
  denom <- exp(-m) + sum_exp

  U_til <- m + log(denom)
  P_til <- sweep(S, 2, exp_fixed, `*`)
  P_til <- P_til / denom
```

```
  loc_sum <- D %*% t(S)
  v <- matrix(U_til, nrow = N, ncol = K, byrow = TRUE) + gamma * loc_sum

  vm <- pmax(0, apply(v, 1, max))
  exp_v <- exp(v - vm)
  denom_search <- exp(-vm) + rowSums(exp_v)
  P_search <- exp_v / denom_search

  P_choice <- P_search %*% P_til
  colMeans(P_choice)
}

con_map_endog <- function(gamma, delta_start, df, loc_list, S,
                          idx_vec, tol = 1e-10, max_iter = 1000) {

  delta <- delta_start
  s_obs <- df$s
  markets <- names(idx_vec)

  for (i in seq_len(max_iter)) {
    s_hat <- numeric(length(delta))
    for (m in markets){
      idx <- idx_vec[[m]]
      Dm <- loc_list[[m]]
      s_hat[idx] <- predict_share_endog(delta[idx], gamma, Dm, S)
    }
    delta_new <- delta + log(s_obs) - log(pmax(s_hat, 1e-12))
    diff <- max(abs(delta_new - delta))
    delta <- delta_new
  }
  list(delta = delta, converged = FALSE, iter = max_iter, diff = diff)
}

delta_last <- delta_0

gmm_obj_endog <- function(theta) {
  beta0 <- theta[1]
  beta1 <- theta[2]
  alpha <- theta[3]
  gamma <- theta[4]

  cm_endog <- con_map_endog(
```

```r
    gamma       = gamma,
    delta_start = delta_last,
    df          = q5_df,
    loc_list    = loc_list,
    S           = S,
    idx_vec     = idx_vec,
    tol         = 1e-12,
    max_iter    = 1000
  )

  delta_last <<- cm_endog$delta
  xi <- cm_endog$delta - (beta0 + beta1 * q5_df$x + alpha * q5_df$p)
  Q  <- crossprod(Z, xi) / n
  as.numeric(t(Q) %*% W %*% Q)
}

theta_start_5 <- c(beta0 = 1, beta1 = 0.2, alpha = -0.25, gamma = -0.1)
if (exists("theta_hat4")) {
  theta_start_5[c("beta0", "beta1",
                  "alpha")] <- theta_hat4[c("beta0", "beta1", "alpha")]
}
```

```r
set.seed(0219)
gmm_opt_endog <- optim(
  par = theta_start_5,
  fn = gmm_obj_endog,
  method = "Nelder-Mead",
  control = list(maxit = 5000, reltol = 1e-8)
)

theta_hat5 <- gmm_opt_endog$par
theta_hat5
gmm_opt_endog$value
gmm_opt_endog$convergence

resolve_endog <- con_map_endog(
  gamma       = theta_hat5["gamma"],
  delta_start = delta_0,
  df          = q5_df,
  loc_list    = loc_list,
  S           = S,
  idx_vec     = idx_vec,
```

```
  tol         = 1e-10,
  max_iter    = 1000
)

est_tab5 <- tibble(
  term = c("Intercept", "Rating Coefficient", "Price Coefficient", "Gamma"),
  estimate = c(theta_hat5["beta0"],
               theta_hat5["beta1"],
               theta_hat5["alpha"],
               theta_hat5["gamma"]),
  true = c(1, 0.2, -0.25, -0.1)
)
kable(est_tab5, digits = 3,
      caption = "GMM Estimates with Endogenous Search vs. True Values")
```